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Research paper

Multi-modality medical image fusion based on image decomposition framework and nonsubsampled shearlet transform

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ABSTRACT

Medical image fusion increases accuracy of clinical diagnosis and analysis through integrating complementary information of multi-modality medical images. A novel multi-modality medical image fusion algorithm exploiting a moving frame based decomposition framework (MFDF) and the nonsubsampled shearlet transform (NSST) is proposed. The MFDF is applied to decompose source images into texture components and approximation components. Maximum selection fusion rule is employed to fuse texture components aimed at transferring salient gradient information to the fused image. The approximate components are merged using NSST. Finally, a components synthesis process is adopted to produce the fused image. Experimental results verify that the proposed method achieves better performance than other compared state-of-art methods in both visual effects and objective criteria.

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1. Introduction

The modern medical imaging technologies provide diversified modalities such as computed tomography (CT), magnetic resonance imaging (MRI), single-photon emission computed tomography (SPECT) and etc., to ensure accuracy of clinical diagnosis. Medical image fusion automatically combines complementary information from multiple source images into single informative one, which helps to improve location precision and make unbiased decisions [1]. With the growing appeal of this huge potential research field, various kinds of multi-modality medical image fusion methods have been proposed [2–6].

The mainstream image fusion methods adopt multi-scale decomposition to capture salient features of source images. Generally, the decomposition procedure is completed with multi-scale transform (MST) tools such as multi-scale pyramid transform [7,8], discrete wavelet transform (DWT) [9], stationary wavelet transform (SWT) [10], dual tree complex wavelet transform (DTCWT) [11,12], nonsubsampled contourlet transform (NSCT) [13,14], nonsubsampled shearlet transform (NSST) [15,16], and so on. Among all the prevalent MST tools, the NSST possesses fine characteristics of shift-invariant, high directional sensitivity and relatively lower computational complexity, which makes it an excellent MST tool for medical image fusion. Current contribution of recently proposed NSST based image fusion methods mainly lies in formulating efficient fusion rules. Liu et al. used morphological component analysis to fuse decomposed low-frequency subbands by NSST [17]. For the fusion of low-frequency and high-frequency subbands, fusion rules are respectively formulated based on sum of variation in square and two different features in Ganasala's method [18]. Yin et al. used singular value decomposition method to estimate structure information of image while fusing low-frequency subbands [5]. Considering the dependence of NSST coefficients, a contextual statistical similarity based fusion method is proposed in [19]. Kong proposed a concise and effective improved pulse-coupled neural network to fuse NSST coefficients and obtain good fusion results [20]. In order to avoid blur and contrast reduction of the fused image, feature-motivated adaptive PCNN is proposed to fuse subbands decomposed with NSST [21]. In addition, NSST based medical fusion method is applied to evaluate six color models in the fusion of functional and anatomical images [22].

The conventional fusion methods based on NSST are directly decomposed the source images. Although NSST with suitable fusion rules may achieve good fusion results, utilizing one transform cannot fully capture salient features especially for edge and textures contained in source images. Recently, a moving frame based decomposition framework (MFDF) is proposed to encode local geometry of an image [23]. By using MFDF, an image can be decomposed into a texture component J_1 containing texture and edge features and an approximate component J_3 similar to source image. The MFDF can be used to separate edge information from source images, which is helpful in transferring more salient edge features

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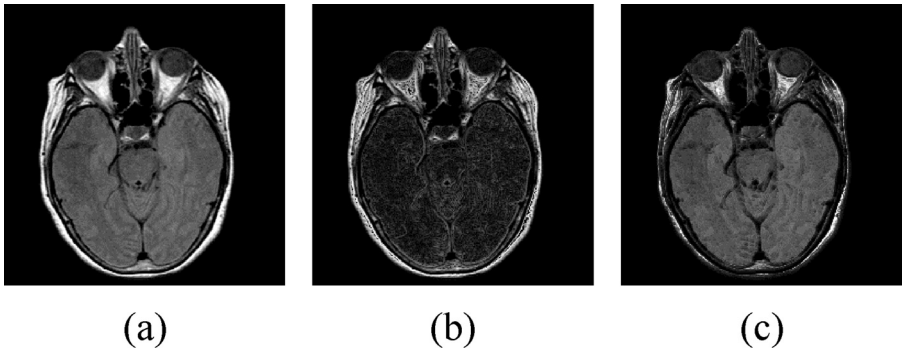

Fig. 1. The decomposition example: (a) an MRI image, (b) component J_1, (c) component J_3.

into the fused image. The separated approximate image including detail information can be extracted using NSST. Through formulating appropriate rules, more information will be transferred from source images into fused image. Out of the above considerations and transfer more edge and texture features of the source image into the fused image, a novel medical image fusion method combining the MFDF and NSST is proposed in this paper. Each source image is firstly decomposed into J_1 and J_3 components using MFDF. As J_1 components mainly maintain edges and textures information, so maximum selection fusion rule is employed to get the fused J_1. Since many significance features are still contained in J_3 components, NSST is used to further decompose the J_3 component to capture and transfer more salient features into the fused image. The average fusion rule and sum-modified-Laplacian (SML) fusion strategy are respectively applied in the fusion of decomposed low-frequency and high frequency subbands of NSST decomposed components. The final fusion result is obtained through synthesising fused J_1 and J_3 components. Experimental results demonstrate that the proposed method achieves a remarkable superiority in both visual effects and objective criteria.

2. Basic theory

2.1. Image decomposition framework

The literature [23] introduces an image decomposition model named as MFDF. It uses a moving frame that encodes local geometry information including gradient's direction and level lines. For an image I, the orthonormal moving frame ($\mathbf{Z_1}$,$\mathbf{Z_2}$,$\mathbf{N}$) is constructed by the following formulas,

$$\mathbf{Z_i} = \frac{\mathrm{d}\psi(z_i)}{\|\mathrm{d}\psi(z_i)\|_2}, i = 1, 2 \tag{1}$$

$$\psi : (x, y) \mapsto (x, y, \mu I(x, y)) \tag{2}$$

where $z_1 = \left(\mu I_x, \mu I_y\right)^{\mathrm{T}}$ and $z_2 = \left(-\mu I_y, \mu I_x\right)^{\mathrm{T}}$ respectively denotes the gradient and level lines of scaled version $\mu I(\mu \in [0, 1])$ of image I. The unit normal $\mathbf{N}$ is orthogonal with $\mathbf{Z}_1$ and $\mathbf{Z}_2$. The columns of $\mathbf{Z}_1$, $\mathbf{Z}_2$ and $\mathbf{N}$ constitute a matrix $\mathbf{D}$ as follows,

$$\mathbf{D} = \begin{pmatrix} \frac{I_x}{|\nabla I|\sqrt{\left(1+\mu^2|\nabla I|^2\right)}} & \frac{-I_y}{|\nabla I|} & \frac{-\mu I_x}{\sqrt{\left(1+\mu^2|\nabla I|^2\right)}} \\ \frac{I_y}{|\nabla I|\sqrt{\left(1+\mu^2|\nabla I|^2\right)}} & \frac{I_x}{|\nabla I|} & \frac{-\mu I_y}{\sqrt{\left(1+\mu^2|\nabla I|^2\right)}} \\ \frac{\mu|\nabla I|}{\sqrt{\left(1+\mu^2|\nabla I|^2\right)}} & 0 & \frac{1}{\sqrt{\left(1+\mu^2|\nabla I|^2\right)}} \end{pmatrix} \tag{3}$$

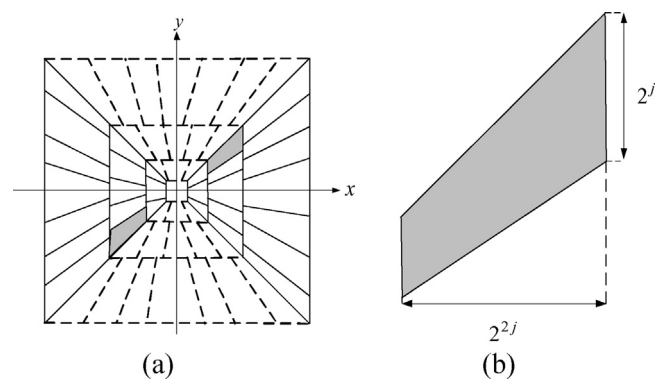

Fig. 2. The (a) frequency plane and (b) frequency support of the frequency partition of NSST [24].

By using the inverse matrix of $\mathbf{D}$, the image I can be decomposed into three components (J_1, J_2, J_3) and decomposition can be expressed as

$$\begin{pmatrix} J_1 \\ J_2 \\ J_3 \end{pmatrix} = \mathbf{D}^{-1} \begin{pmatrix} 0 \\ 0 \\ I \end{pmatrix}. \tag{4}$$

As Eq. (4) indicates that the component J_2 is always $\mathbf{0}$. The decomposed texture component J_1 mainly contains edges and textures, and the approximate component J_3 is the difference between the original image and its gradient's norm. Fig. 1 shows decomposition results of an MRI image, where the parameter μ is set to 0.05.

2.2. Nonsubsampled shearlet transform

As a multi-scale geometric analysis tool, the NSST has shift-invariance property and anisotropic direction selectivity. Compared to the discrete wavelet transform which is only good at capturing point-wise singularities, the NSST is a true 2-D sparse representation tool for its excellent performance in detecting linear singularities. The NSST uses nonsubsampled Laplacian pyramid filter to implement multi-scale partition and shearing filters to complete directional localization. Besides the shift-invariance and flexible directional selectivity, it has higher computational efficiency than NSCT. A schematic diagram of frequency partition of NSST is shown in Fig. 2, from which we can see that the support is a pair of trapezoids with size $2^{2j} \times 2^j$. More details about NSST can be found in literature [24].

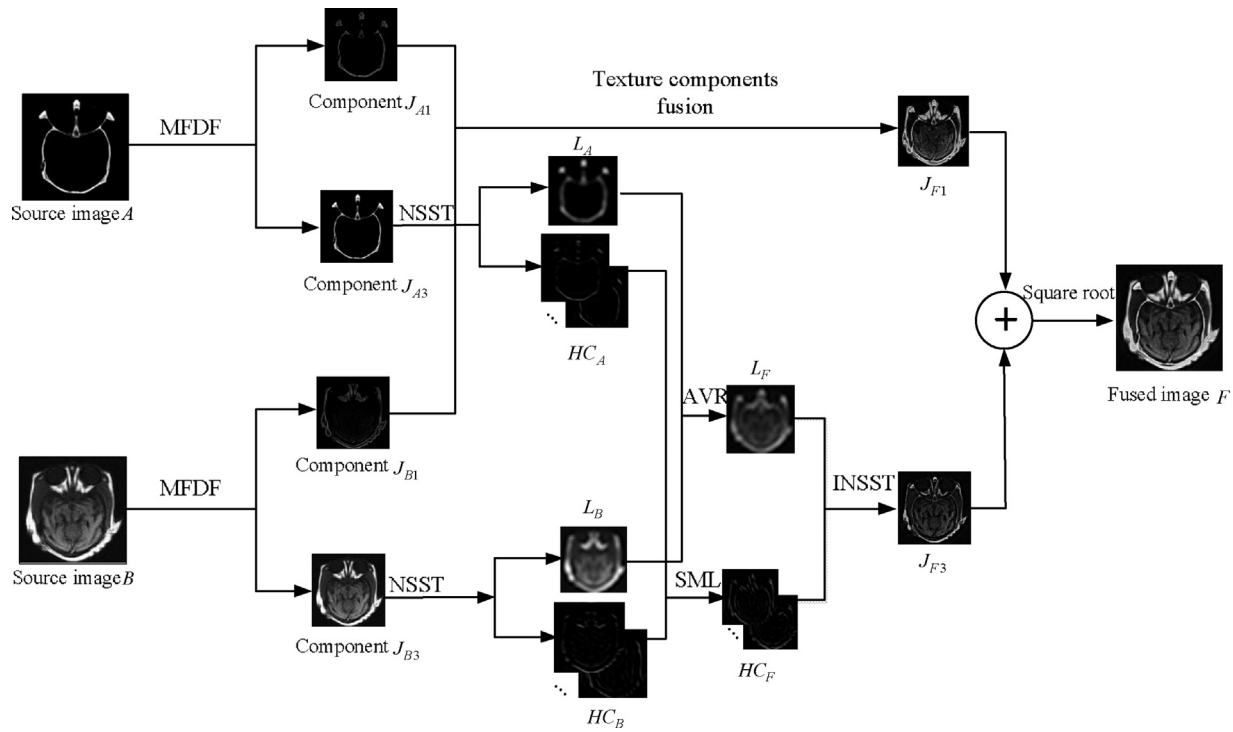

Fig. 3. The flowchart of decomposition framework and NSST based medical image fusion algorithm.

3. Fusion method for multi-modality medical images

Image registration is necessary before fusion operation to ensure the structures of multimodality medical images could be matched. For two multi-modality medical images A and B, they have been perfectly matched by registration methods [25]. The MFDF is applied to decompose each image. The decomposed texture component and approximate component of image A are respectively denoted as J_{A1} and J_{A3}, and for image B are respectively denoted as J_{B1} and J_{B3}. The details of the proposed image fusion method using are elaborated in the following sections.

3.1. Texture components fusion

Considering human vision system is highly sensitive to the texture component J_1, maximum selection rule is used for the fusion of J_{A1} and J_{B1} to ensure that more salient edge or texture features can be transferred into the fused image. The corresponding pixels of J_{A1} and J_{B1} with lager values are selected as the fused texture component J_{F1}. The fusion rule is expressed as

$$J_{F1} = \begin{cases} J_{A1} \text{if} J_{A1}(i,j) > J_{B1}(i,j) \\ J_{B1} \text{if} J_{A1}(i,j) \le J_{B1}(i,j) \end{cases}, \tag{5}$$

where (i, j) denotes the pixel position of texture component.

3.2. Approximate components fusion

The approximate component J_3 is similar to the source image, which retains most features of original image and only the gradient's norm is subtracted. In order to transfer more salient features from approximate component into fused image, NSST is employed in the fusion process.

After the approximate components J_{A3} and J_{B3} respectively are decomposed by NSST, one low frequency subband and a series of high frequency subbands are obtained. The low frequency subband contains main energy of approximate component, so average fusion rule (AVR) is used to reduce distortion of fused image. The fused low frequency subband L_F is computed by taking average pixel value of the two low frequency subbands of J_{A3} and J_{B3},

$$L_F(i,j) = \frac{L_A(i,j) + L_B(i,j)}{2}, \tag{6}$$

where (i, j) denotes the pixel position of low frequency subband. The L_A and L_B denote the decomposed low frequency subbands of J_{A3} and J_{B3} components, respectively.

Since detail information mainly exists in the directional high frequency subbands, the SML [26] which can well reflect salient features is used as activity level measure to select high frequency coefficients. The SML is computed in a $(2M+1) \times (2M+1)$ size window, which centered at (i, j) and the calculation formula is defined as

$$SML(i,j) = \sum_{k=-M}^{M} \sum_{l=-M}^{M} [ML(i+k, j+l)]^2, \tag{7}$$

$$\begin{aligned} ML(i,j) = &|2HC(i,j) - HC(i-1,j) - HC(i+1,j)| \\ &+|2HC(i,j) - HC(i,j-1) - HC(i,j+1)| \end{aligned}, \tag{8}$$

where $HC(i, j)$ denotes high frequency coefficient at pixel (i, j). If the parameter M is too large, the SML will not exactly reflect the salient features. Generally, M is set to 1 and the algorithm gives the best results. For the obtained high frequency subbands, the coefficient with maximum SML value at pixel (i, j) will be selected as the fused coefficient. The fusion rule of high frequency subbands is

$$HC_F^{l,k}(i,j) = \begin{cases} HC_A^{l,k}(i,j) \text{if} SML_A^{l,k}(i,j) > SML_B^{l,k}(i,j) \\ HC_B^{l,k}(i,j) \text{if} SML_A^{l,k}(i,j) \le SML_B^{l,k}(i,j) \end{cases}, \tag{9}$$

where $HC^{l,k}(i, j)$ denote coefficient at pixel (i, j) of l-th scale and k-th direction subband.

After obtaining the fused low frequency subband L_F and fused high frequency subband $HC_F^{l,k}(i, j)$, the fused approximate component J_{F3} is achieved by performing inverse NSST (INSST).

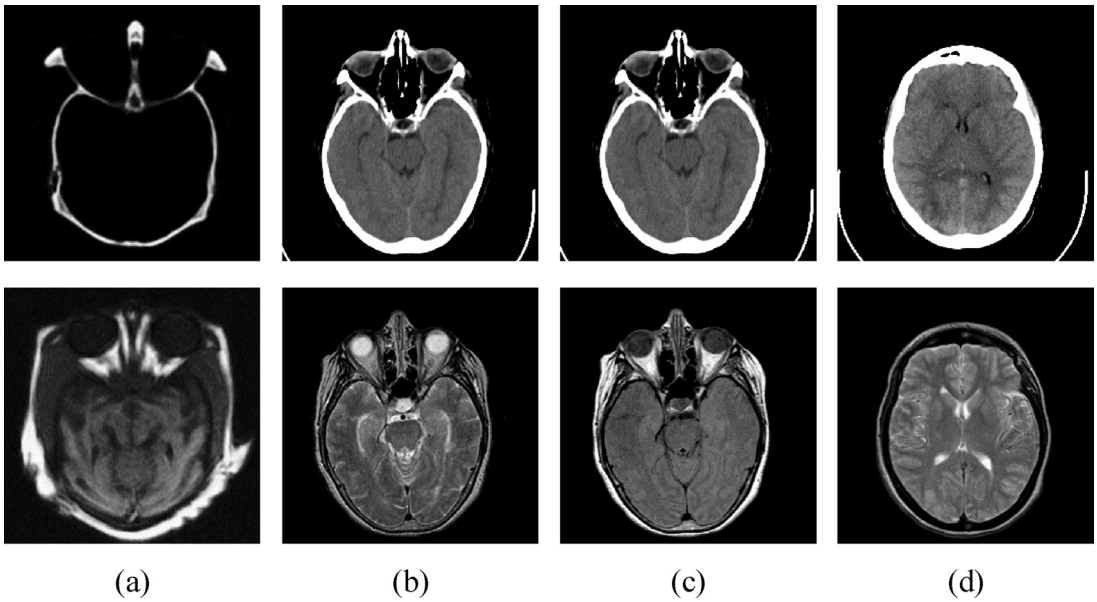

Fig. 4. The source medical image sets used in the experiments, namely (a) Med-1, (b) Med-2, (c) Med-3 and (d) Med-4.

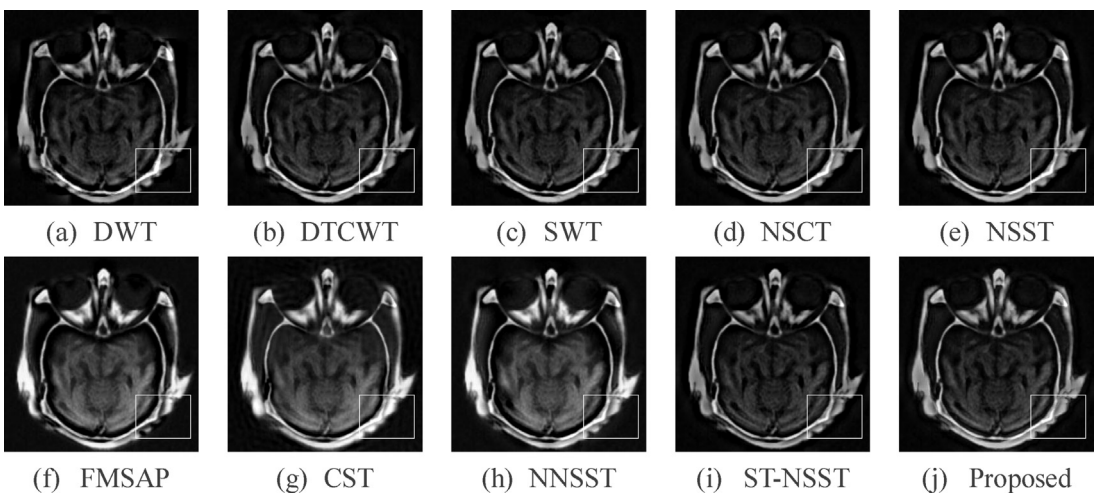

Fig. 5. Fusion results obtained from different methods on the "Med-1" image set.

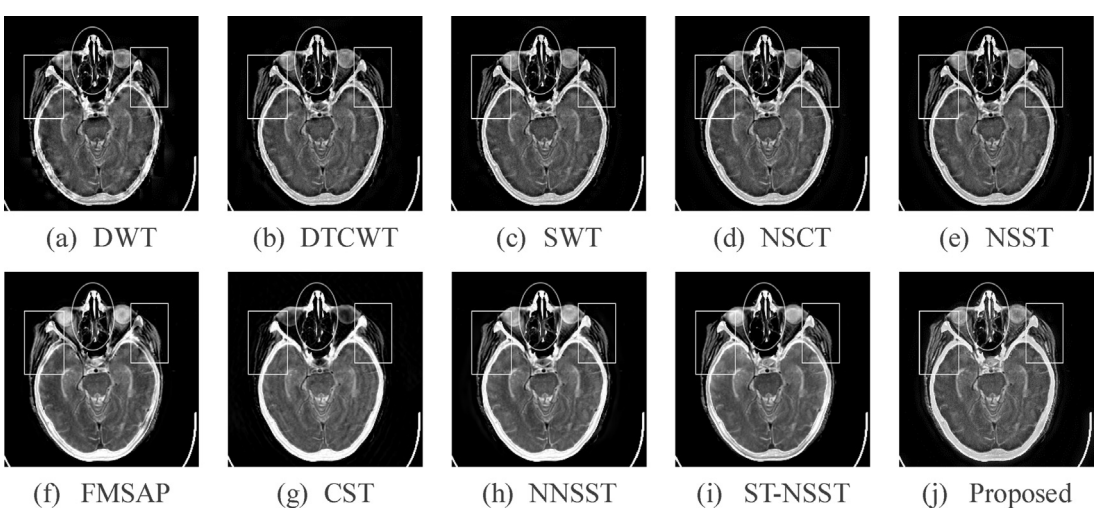

Fig. 6. Fusion results obtained from different methods on the "Med-2" image set.

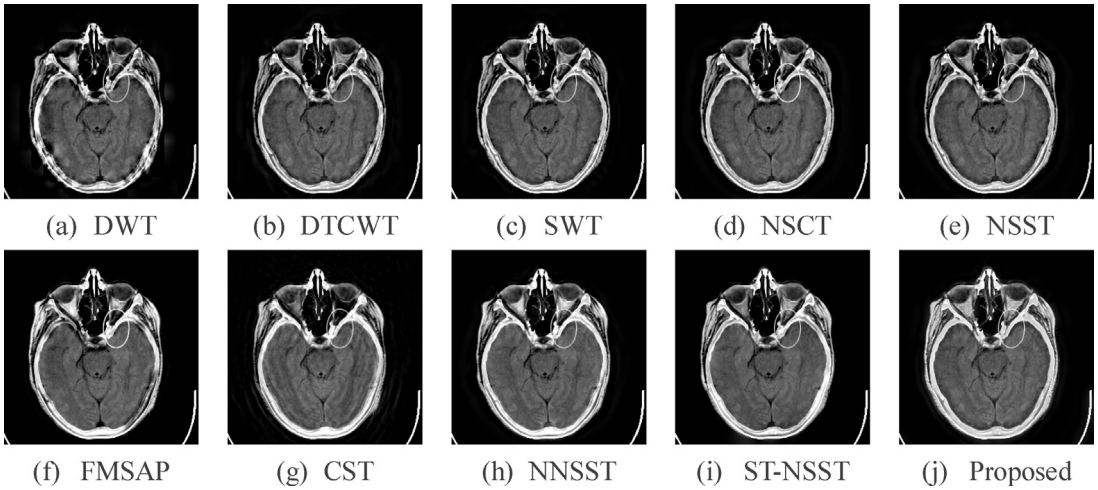

Fig. 7. Fusion results obtained from different methods on the "Med-3" image set.

3.3. Components synthesis process

As the decomposition framework is completed with multiplying matrices and the sum of decomposed components is not equal to original image, so summing the J_{F1} and J_{F3} directly won't lead to good fusion result. Noticed that the sum of the squares of decomposed texture component J_1 and approximate component J_3 is exactly the square of original image I, so the final fused image F is produced through components synthesis process, i.e., calculating square root of J_{F1}^2 and J_{F3}^2,

$$F = \sqrt{J_{F1}^2 + J_{F3}^2}. \tag{10}$$

The implementation procedure of the proposed multi-modality medical image fusion algorithm is illustrated in Fig. 3 and the general fusion process can be summarized as follows:

(1) Using image decomposition framework, the source images are respectively decomposed into two components.
(2) The maximum selection fusion rule is employed in the fusion of texture components and the approximate components are fused using NSST decomposition.
(3) The final fused image F is obtained by synthesizing the fused texture component and fused approximate component.

4. Experimental results and discussions

In order to intuitively demonstrate the superiority of the proposed multi-modality medical image fusion algorithm, the fusion results obtained from the proposed method and five other state-of-the-art methods including DWT [9], SWT [10], DTCWT [12], NSCT [27], NSST [28], feature-motivated simplified adaptive PCNN (FMSAP)[21], complex shearlet (CST) [29], NNSST[5] and ST-NSST [30] based methods are put together for comparison. The size of source images used in the experiments is 256×256. The decomposition level for CST method is set as 3 and other multi-resolution based methods are set as 4. The number of directions in NSCT and NSST decomposition is set as (4, 4, 8, 8) from coarser to finer levels. The other parameters are set as the default values given by the corresponding references. Four source image sets are shown in Fig. 4, which are collected from Imagefusion.org [31] and the whole brain atlas website [32]. For each column, the CT image is shown above the corresponding MRI image. It can be seen that the CT images clearly show bone structures while soft issue structures are accu-

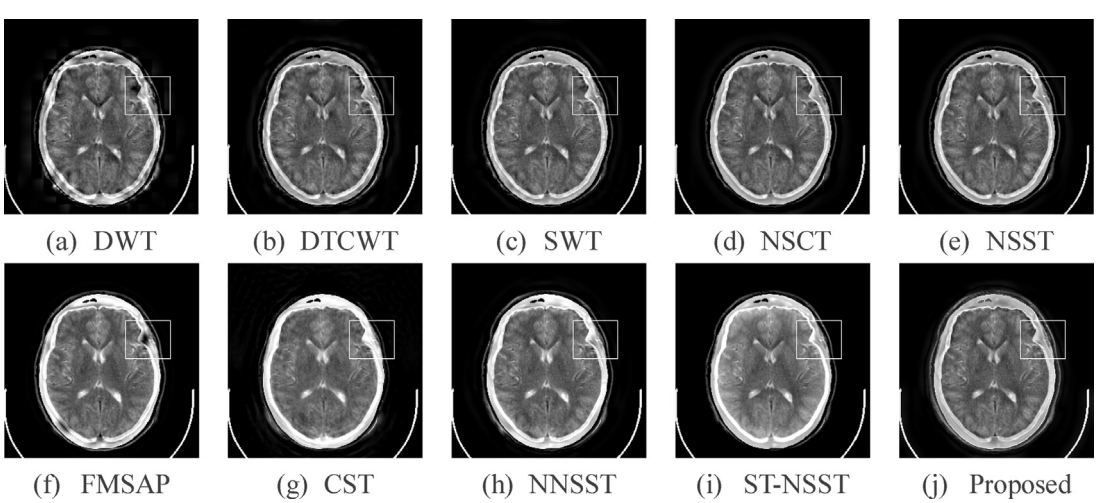

Fig. 8. Fusion results obtained from different methods on the "Med-4" image set.

Table 1
The quantitative assessment results of different fusion methods.

Images	Metric	DWT	DTCWT	SWT	NSCT	NSST	FMSAP	CST	NNSST	ST-NSST	Proposed
Med-1	MI	2.1141	2.1044	2.3361	2.2087	2.2008	**3.0698**	2.5508	2.8609	2.2059	2.3761
	Q_E	0.4290	0.4473	0.4794	0.4761	0.4688	0.4989	0.3285	0.4811	**0.5792**	0.5120
	Entropy	6.1512	6.2074	6.1063	6.1488	6.1706	6.4152	6.2724	6.2575	6.2174	**6.4179**
Med-2	MI	2.2169	2.3372	2.4355	2.4224	2.3919	2.7044	2.5462	2.6060	2.3999	**2.7617**
	Q_E	0.2085	0.2343	0.2503	0.2474	0.2475	0.2462	0.1929	0.2597	0.2447	**0.2645**
	Entropy	5.1884	5.3264	5.1447	5.2680	5.2291	4.7608	5.3206	5.0938	5.1667	**5.9679**
Med-3	MI	2.1713	2.3357	2.4453	2.4225	2.4082	2.6215	2.4048	2.5596	2.4117	**2.7093**
	Q_E	0.1698	0.1947	0.2010	0.2017	0.2042	0.2076	0.1463	0.2228	0.2278	**0.2319**
	Entropy	5.1951	5.3609	5.1538	5.2804	5.2617	4.6913	5.6081	5.1542	5.1361	**5.9869**
Med-4	MI	2.2868	2.3791	2.4828	2.5046	2.4658	2.6223	2.6248	2.5658	2.4268	**2.6250**
	Q_E	0.2070	0.2243	0.2487	0.2489	0.2473	0.2444	0.1708	0.2469	**0.2596**	0.2535
	Entropy	5.3774	5.4233	5.2668	5.3324	5.2941	4.6362	5.6273	5.1693	5.2237	**6.0949**

rately pictured in MRI. The desirable fusion results should contain salient features extracted from both CT and MRI images.

4.1. Fusion results of medical images

The comparisons of fusion results obtained from different methods are demonstrated in Figs. 5–8. In general, the quality of fused image is evaluated with visual observation and objective evaluation. Both of them are used to evaluate fused images in our experiments.

4.1.1. Visual observation

Fig. 5(a)–(f) show the fusion result of "Med-1" image set, which is shown in Fig. 4(a). The source CT image well describes bone structures and source MRI image emphasizes on soft tissues. It can be seen that main features of CT and MRI images are preserved in the fused images by all methods, the differences among them marked with rectangular are still obviously exist. The fused image (a) introduced some inconsistency since DWT only well capture pointwise singularities. Varying degrees of artifacts are also appearing in fused images (b) and (c). The contrast demonstrated in images (f)-(h) is better than other methods, but some detail information is lost especially in overlapping parts of CT and MRI images. The contrast of Fig. 5(i) is worse than the fused image (j). The fusion result (j) obtained from the proposed method has a better visual effect in the aspects of contrast and edge details than other compared method, the bone structure can be easily distinguished from soft issues.

The fusion results of "Med-2" using different methods are presented in Fig. 6. The main differences of the fused images are marked with rectangles and ovals. More details especially edges are displayed in the fused image (j). In addition, the key soft issues are reserved the most and the edges and contours are sharper in (j) than in other images.

The similar performance is verified in Fig. 7, which shows the fusion results of "Med-3" image set. The ovals point out the advantages of the proposed fusion method. The fused images shown in Fig. 6(d) and (e) introduce some artifacts in the edge region. Compared with other methods, Fig. 7(j) gives less distinction or less contrast in some superposed positions. However, it retains the most detail and edge information of the source images and the main features are easily distinguished, which is superior to other competitive methods.

The fusion example on the "Med-4" image set is shown in Fig. 8, from the marked positions can be seen that the proposed method appropriately enhances the edges and gives a better vision compared with the fusion results obtained from DWT, DTCWT, SWT, NSCT, FMSAP, CST, NNSST and ST-NSST methods.

4.1.2. Objective evaluation

In some cases, to judge a fused image is difficult only through visual perception. Therefore, objective evaluation criteria are necessary used to evaluate the quality of fused images. The main advantage of the proposed method is retaining more edge and detail information, so three metrics are selected to reflect the amount of information transferred from source images to the fused image. Three evaluation metrics including edge-dependent fusion quality index Q_E [33], mutual information (MI) [34] and entropy [35] are considered as objective criteria.

The Q_E computes the edge information transferred into the fused image from the source images. The edge-dependent fusion quality index Q_E is defined as:

$$Q_E(A,B,F) = Q_W(A,B,F)\cdot Q_W\left(A',B',F'\right) \tag{11}$$

where the symbol A', B' and F' are edge images corresponding to A, B and F, respectively. The Q_W denotes weighted fusion quality index, and the definition is based on sliding window technique [36],

$$Q_W(A,B,F) = \sum_{\omega\in W} c(\omega)\left[\lambda(\omega)Q(A,F|\omega) + (1-\lambda(\omega))Q(B,F|\omega)\right] \tag{12}$$

where $c(\omega)$ and $\lambda(\omega)$ are weight factor calculated with window feature. The $c(\omega)$ and $\lambda(\omega)$ are defined as:

$$c(\omega) = \frac{C(\omega)}{\sum_{\omega\in W} C(\omega')},\ C(\omega) = \max\left(s(A|\omega), s(B|\omega)\right) \tag{13}$$

$$\lambda(\omega) = \frac{s(A|\omega)}{s(A|\omega)+s(B|\omega)} \tag{14}$$

where $s(A|\omega)$ and $s(B|\omega)$ denote the activity level in window ω of image A and B, respectively.

The MI measures the amount of information transferred into the fused image and the entropy reflects the information of fused image. The information entropy is defined as:

$$\text{Entropy} = -\sum_{i=0}^{L-1} P_i \log_2(P_i) \tag{15}$$

where P_i denotes the probability of pixels which value are i. The symbol L denotes the number of gray levels. The MI is defined as:

$$MI((A,B),F) = \sum_{i=0}^{L-1}\sum_{j=0}^{L-1}\sum_{k=0}^{L-1} P_{ABF}(i,j,k)\log_2\left(\frac{P_{ABF}(i,j,k)}{P_{AB}(i,j)P_F(k)}\right) \tag{16}$$

where $P_{AB}(i,j)$ denotes joint probability of image A and B, and $P_{ABF}(i,j,k)$ is joint probability of image A, B and F.

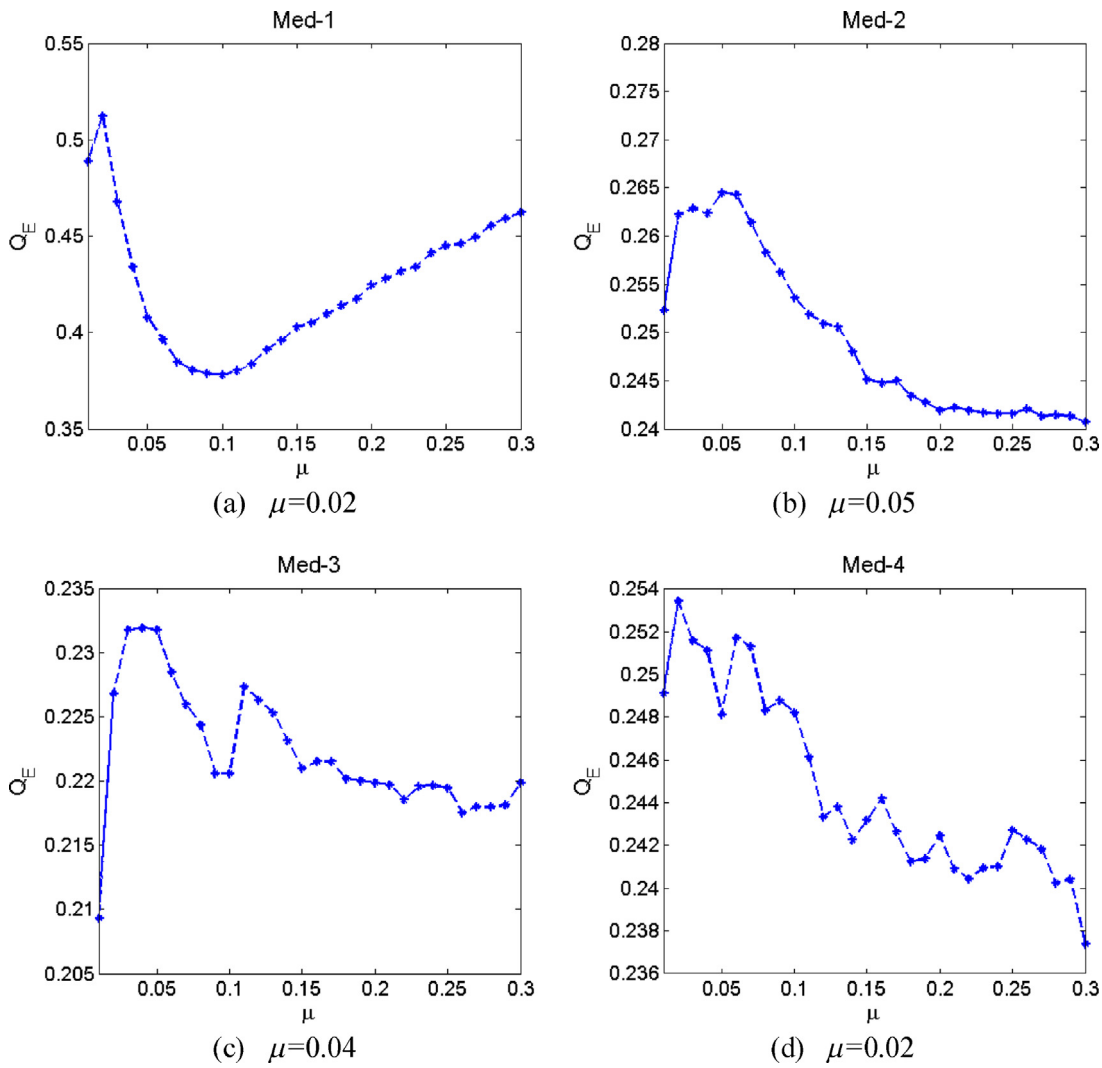

Fig. 9. The Q_E curves versus different μ values of four medical image sets.

For the three objective evaluation criteria, the larger values mean better results.

Table 1 shows the objective quantitative assessments of different methods and the best values are highlighted in bold. For the "Med-1" image set, although the values of MI and Q_E are respectively lower than FMSAP method and ST-NSST method, the fusion result of proposed method shown in Fig. 5 is better than the two methods. For the "Med-4" image set, there is not much difference of Q_E between the proposed method and ST-NSST method, but the values of MI and entropy are higher. For the other two image sets, the values of three metrics are all higher than others, which mean that the proposed method obtains a better fusion result and transfers more information from source images to the fused image.

4.2. Parameter setting

In the whole image fusion process, only the parameter μ needs to be adjusted manually. If the value of μ is set too large, some edge and texture details will lost in the fusion of texture components. On the contrary, the edge and texture details will not be highlighted in the fused image if the value of μ is set too small. Therefore, set an appropriate value for μ is very important. As the objective evaluation metric named edge-dependent fusion quality index Q_E is valid in evaluating the fusion performance by taking into account the importance of edge information, so the value of Q_E is used to guide parameter setting of μ. Fig. 9 shows the Q_E curves under continuous variations of μ from 0.01 to 0.3. It can be seen that the value of Q_E is large when μ changed in the range of 0.02-0.05. For different image sets, the experiments with different μ will perform 4 times and the result with higher value of Q_E will be selected as the final fusion result. For each image set, the recommended value of μ is given under corresponding curves.

5. Conclusion

In this paper, a novel image fusion algorithm based on an image decomposition framework MFDF and NSST is proposed for fusion of the CT and MRI images. Unlike conventional multi-scale transform based methods, the proposed algorithm additionally utilizes an image decomposition framework before NSST decomposition to capture gradient information. Therefore, more salient edge features and detail information can be transferred into the fused image. Experimental results demonstrate that the performance of the proposed method superior to other multi-scale fusion method in both visual effect and objective assessment.

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